Chapter 7, Problem 6RQ

Problem

In the circuit in Fig. 7.79, $v(\infty)$ is:

(a) 10 V

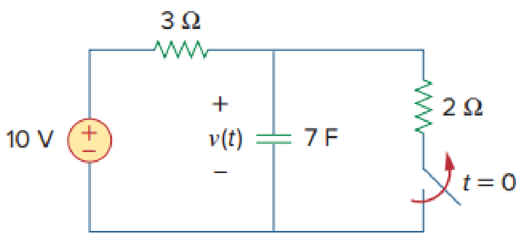
(b) 7 V

(c) 6 V

(d) 4 V

(e) 0 V

Fig. 7.79:

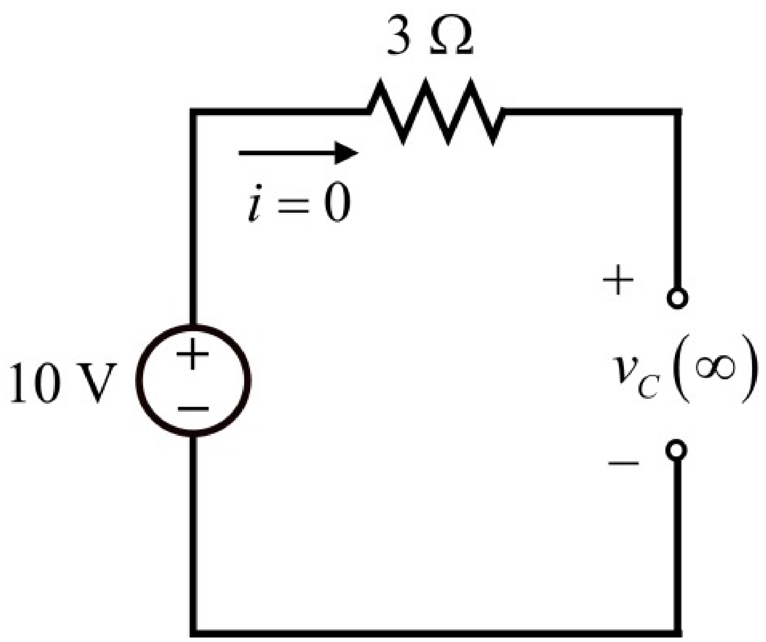


Step-by-step solution

Step 1 of 3

Refer to Figure 7.79 in the text book.

At $t = 0$, the switch is open. At $t = \infty$, circuit will reach to steady state and at steady state capacitor acts like an open circuit. Draw the modified circuit diagram for $t = \infty$.



Step 2 of 3

Since the capacitor is open, no current flows through the $3\ \Omega$ resistor and hence voltage drop across the resistor is zero. So the complete source voltage of 10 V appears across the open circuit terminals of the capacitor.

Therefore, the voltage across capacitor at $t = \infty$ is $\boxed{10\ \text{V}}$.

Step 3 of 3

Thus, the correct option is $\boxed{(\mathbf{a})}$.